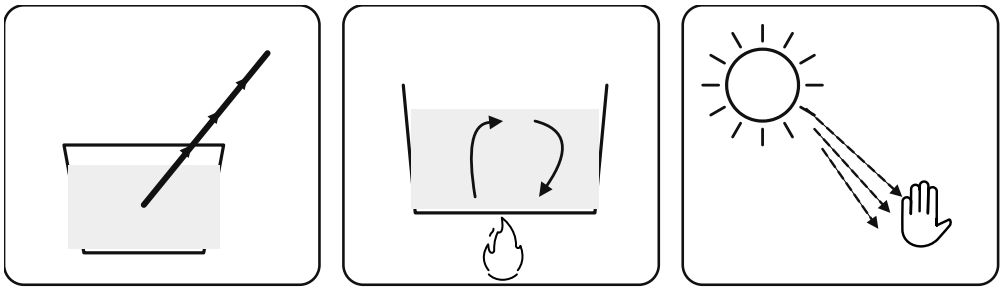
PHYSICAL AND CHEMICAL SCIENCE

Heat and Temperature

Sources of heat, measuring temperature, heating and cooling, conduction, convection and radiation, and heat safety



conduction
heat travels through solids

convection
warm liquid or air rises

radiation
heat travels as rays

In this pack

📖 **Learn it:** Heat and temperature ★☆☆
Fact check: Hot and cold facts ★☆☆
Think it through: Thinking about heat ★★☆
Investigate: Which cover keeps tea hot the longest? ★★☆
Real-world case study: Keeping warm safely in winter ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C3 · HEAT

Heat and temperature

★☆☆ Grade 3

Name: Class: Date:

Learning goals

- Explain what heat is and name sources of heat.
- Describe what happens when heat is added or taken away.
- Explain how heat travels and how to stay safe near hot things.

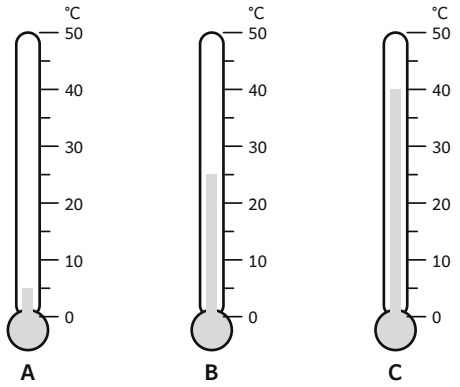
Rub your hands together quickly. They get warm! **Heat** is a kind of **energy** that makes things warmer. **Temperature** tells us how hot or cold something is.

Sources of heat

the Sun	fire	light bulbs	electric heaters and irons	rubbing things together

Measuring temperature

We measure temperature in **degrees Celsius (°C)** with a **thermometer**. The liquid inside rises when it gets warmer. Ice melts at **0 °C**, your body is about **37 °C**, and water boils at **100 °C**.



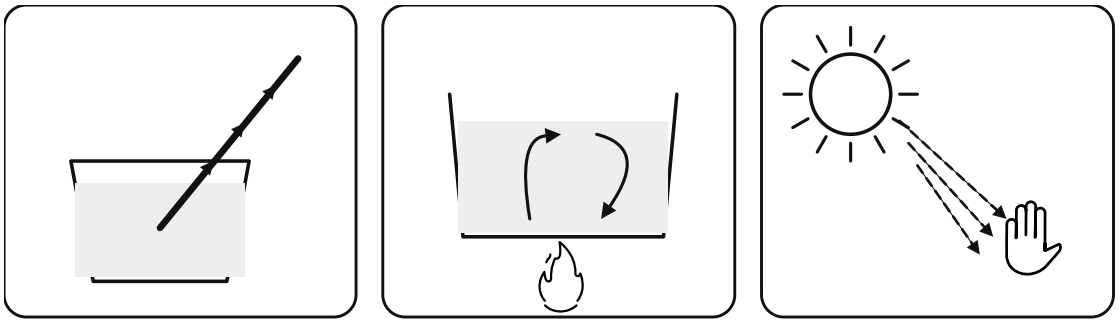
Read the thermometers: A is 5 °C, B is 25 °C, C is 40 °C.

Adding and removing heat

Adding heat	Removing heat (cooling)
ice melts into water	water freezes into ice
water boils and turns to steam	steam turns back into water drops
butter and chocolate melt	melted candle wax goes hard
rice and eggs cook	hot tea gets cool

How heat travels

Heat always moves from **hotter** places to **colder** places, in three ways:



conduction
heat travels through solids

convection
warm liquid or air rises

radiation
heat travels as rays

Three ways heat travels.

- **Conduction**: heat travels through solids. A metal spoon in hot tea gets hot all the way up.
- **Convection**: warm water or air rises and cool water or air sinks. This makes water move around in a pot.
- **Radiation**: heat travels as invisible rays, like the warmth you feel from the Sun or a fire.

Metals let heat through easily: they are good **conductors**. Wood, plastic, cloth and air do not: they are **insulators**. That is why pot handles are wood or plastic, and why a thick kantha quilt keeps you warm in winter.

Using heat safely

We use heat to cook food, boil water to make it safe, iron clothes, dry fish (shutki) and fire clay pots. But heat can be dangerous.

> ⚠ **Stay safe**
>
> Keep away from fires, stoves, hot oil, irons and steam. Use oven gloves. Tie back loose clothes near a fire. If you get burnt, cool the burn under clean running water for 20 minutes and tell an adult.

Key words

heat	energy that makes things warmer
temperature	how hot or cold something is
thermometer	measures temperature
conductor	lets heat through easily
insulator	does not let heat through easily

FACT CHECK · C3 · HEAT

Hot and cold facts

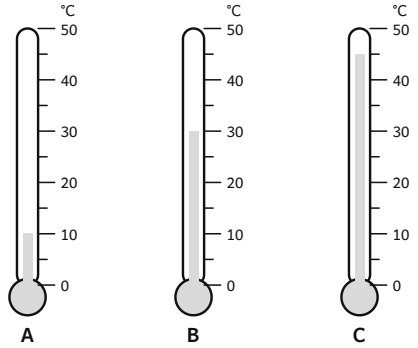
★☆☆
Grade 3

Name: Class: Date:

Learning goals

- Read thermometers.
- Sort changes.

1 Read the thermometers. [3]



A	B	C

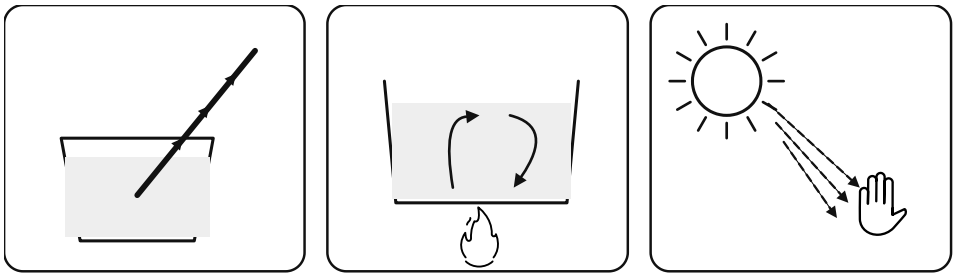
2 Is heat added or removed? [6]

SORT THESE

rice cooks **ice cream melts** **water freezes in the freezer**
melted wax goes hard **wet clothes dry in the sun** **hot tea cools**

Heat added	Heat removed

3 **Name the way heat travels in each picture.** [3]



A B C

A	B	C

THINK IT THROUGH · C3 · HEAT

Thinking about heat

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Explain conductors and insulators.

1 **Why do cooking pots have wooden or plastic handles?** [2]

2 **Why do we wear several layers of clothes in winter?** [2]

3 **Sort the materials.** [5]

SORT THESE

steel spoon **aluminium pot** **wool** **wooden spoon** **plastic handle**

Conductor	Insulator

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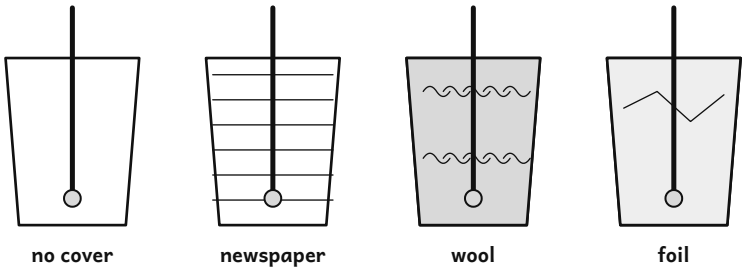
Which cover keeps tea hot the longest?

★★☆ Grade 3

Name: **Class:** **Date:**

Learning goals

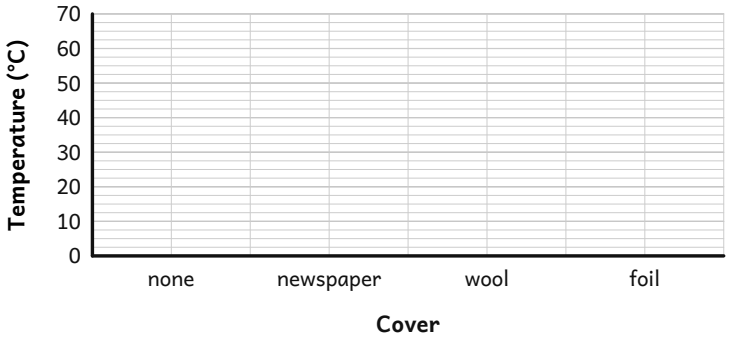
- Measure temperature.
- Draw a bar chart and conclude.



Class 3A filled four cups with hot water at **80 °C**. They wrapped three cups in different materials. After 20 minutes they measured the temperature. (Only an adult should pour the hot water.)

1 **Draw a bar chart.** [3]

Cover	none	newspaper	wool	foil
Temperature after 20 min (°C)	45	52	60	55



2 **Which cover was the best insulator?** [1]

.....

3 **Why did the cup with no cover cool fastest?** [2]

.....

.....

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REAL-WORLD CASE STUDY · C3 · HEAT

Keeping warm safely in winter

★★★
Grade 3

Name: _____ **Class:** _____ **Date:** _____

Learning goals

- Use what you know about heat to stay safe.

Science | Health | Community

In winter, nights in the north of Bangladesh can get very cold, below **10 °C**. Many families sit around small open fires to keep warm.

Every winter, some people are badly burnt when loose clothes such as a sari, orna or shawl catch fire. Doctors and fire fighters teach families how to keep warm safely: wear warm layers and use quilts, sit at a safe distance from fires, tie back loose clothes, and never leave a fire alone.

Based on winter safety advice in Bangladesh.

1 **How does a fire warm people who are sitting near it?** [1]

.....

2 **Write three safety rules for sitting near a fire.** [3]

.....

.....

.....

3 **If someone's clothes catch fire, they should stop, drop and roll. Draw the three steps.** [3]

Stop, drop, roll

REVIEW AND CHALLENGE · C3 · HEAT

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use heat words.

1 Find the words.

G	O	F	C	O	O	L	G	H
H	Y	P	W	D	G	R	J	Y
D	S	U	N	U	R	U	S	L
N	N	W	A	L	T	D	C	D
J	B	K	W	H	M	E	L	T
B	H	F	R	E	E	Z	E	P
O	J	F	I	R	E	F	Y	H
I	C	G	H	E	A	T	K	G
L	G	G	T	F	D	N	P	E

boil cool fire freeze heat melt sun

2 Temperature is measured with *a*... [1]

◯ ruler ◯ thermometer

◯ magnet

3 Heat travelling up *a* metal spoon is... [1]

◯ conduction ◯ convection

◯ radiation

4 Challenge. Design *a* box to keep *a* lunch of hot rice warm. What materials would you use? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name sources of heat	○	○	○
read a thermometer	○	○	○
say what heating and cooling do	○	○	○
explain conduction, convection and radiation	○	○	○
stay safe near hot things	○	○	○